1. Your company is launching a new line of "Smart Office" gear by producing three products:
   - Smart Lamps: Profit = $30 per unit
   - Wireless Chargers: Profit = $45 per unit
   - Digital Organizers: Profit = $20 per unit

   There are two critical resources that limit your production for the month:
   - Assembly Time: 1,200 total hours available.
   - Specialized Sensors: 600 units available.

Product	Assembly (Hours)	Sensors (Units)
Smart Lamp	2	2
Wireless Charger	4	1
Digital Organizer	1	1

Formulate a model to maximize the profit.

$$Max\ Z = 30x_1 + 45x_2 + 20x_3$$

Subject to:

$$2x_1 + 4x_2 + 1x_3 \leq 1{,}200$$
$$1x_1 + 1x_2 + 1x_3 \leq 600$$
$$x_1, x_2, x_3 \geq 0$$

Solution:

$$Z = 17{,}000$$

$$x_1 = 0$$
$$x_2 = 200$$
$$x_3 = 400$$

2. A large manufacturer is interested in your factory and wants to rent your entire production capacity for the month. They want to minimize the total amount they pay you to "buy out" your capacity. They want to find the **lowest possible prices for labor and sensors** that you will actually accept. You will only agree to their prices if their offer is at least as high as the profit you could have made by using those resources yourself. Write an LP model to minimize the total buyout of the resources for the manufacturer.

$$\min W = 1{,}200y_1 + 600y_2$$

Subject to:

$$2y_1 + 2y_2 \geq 30$$
$$4y_1 + 1y_2 \geq 45$$
$$1y_1 + 1y_2 \geq 20$$
$$y_1, y_2 \geq 0$$

Solution:

$$W = 17{,}000$$

$$y_1 = 8.33$$
$$y_2 = 11.33$$